SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Main1.java
Fruits : [Apple, Banana, Cherry]
New Fruits : [Apple, Cherry]
Size : 2
Contains Apple? : true
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Main2.java
Vector : [10, 20, 30]
New Vector : [20, 30]
Size : 2
Element : 20
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Main3.java
Original String : Programming
After Append    : Programming in Java
After Insert    : Programming  Languagein Java
After Replace   : Coding  Languagein Java
After Delete    : Codingein Java
After Reverse   : avaJ niegnidoC
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Main4.java
Tokens are :
Java
C++
Python
JavaScript
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Exercise/Ex1.java
------ TO DO List ------
1.Study
2.Play a sport
3.Coding
4.Build a project
5.Nap

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp5/Exercise/Ex2.java
Registered Courses
------------------
1. Java
2. Data Structures
3. Operating Systems

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ 
```

Conclusion

This assignment helped in understanding the use of ArrayList, Vector, and StringBuffer in Java for dynamic data storage, management, and string manipulation. It demonstrated how ArrayList and Vector can efficiently store and manage collections of data, while StringBuffer enables efficient modification of strings without creating new objects. The implementation improved my understanding of the Java Collections Framework, dynamic memory management, and mutable string operations. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, database management systems, and other real-world Java applications that require efficient data handling and string processing.